

Specification for PCB: Control_Board_C201.PcbDoc

The 6-layer PCB is intended for manual and automatic soldering of SMT and THT parts on the top and bottom sides .

General requirements

IPC-A-600/6010 class 2

Laminate

Laminate: IPC 4101/21 with minimum Tg=130°C

Flammability: V0 according to UL94

CTI index 3: $175 < CTI \leq 400$ or better

CTE: Pre-Tg. 60ppm/°C Post-Tg. 300ppm/°C

Copper plating

Hole plating: min. 20µm. and min. avg. 25µm according to IPC-6012 class 3

Dimensions and Marking

PCB Size (X,Y): 100x70 mm

Tolerances on outline dimensions: +/- 0.2mm

Layer stackup

The PCB stack is in the picture below. The total thickness of the PCB is 1,22 mm ±10%.



Controlled impedance traces

Controlled impedance traces are as follows.

Layer	Impedance Structure	Trace width, mm	Diff line space, mm	1 ref. plane	2 ref. plane	Target Impedance, Ohm*
1	Edge Coupled Coated Microstrip	0.152	0.152	2	-	90 diff
1	Single Ended	0.152	-	2	-	50
2						
3						
4						
5						
6						

*Tolerance on all impedances is $\pm 10\%$.

Solder mask

Liquid film or electrostatic spray coating on top and bottom sides .

According to: IPC-SM-840E, Class T

Minimum thickness: 8 μ m

Maximum thickness: 30 μ m

Legend:

Silkscreen Colour: White on top and bottom layers.

Plugged via holes:

Via holes covered by solder mask on both sides must be plugged.

Via holes with solder mask opening on one side and covered by solder mask on the other side, must be plugged from the same side they are covered by solder mask – solder mask opening on the other side must remain.

Surface finish

ENIG – Au 0.05 μ m / Ni 3.0 μ m;

Solderability accordingly to IPC J-STD-003, test C1 PTH, SMD test F1 set A

Gerber data (.Manufacturing\Gerber Files)

Filename	Contents
20201215-VSI-AcmeCPU_Control_Board_C201_001_PCB_SPECIFICATION.pdf	This file
20201215-VSI-Acme_CPU_Control_Board_C201_001_Gerber.GTL	Layer 1, 1-SIG-Top
20201215-VSI-Acme_CPU_Control_Board_C201_001_Gerber.GP1	Layer 2, 2-GND
20201215-VSI-Acme_CPU_Control_Board_C201_001_Gerber.GP2	Layer 3, 3-PWR
20201215-VSI-Acme_CPU_Control_Board_C201_001_Gerber.G1	Layer 4, 4-SIG-HOR
20201215-VSI-Acme_CPU_Control_Board_C201_001_Gerber.G2	Layer 5, 5-SIG-VERT
20201215-VSI-Acme_CPU_Control_Board_C201_001_Gerber.GBL	Layer 6, 6-GND-BOT
20201215-VSI-Acme_CPU_Control_Board_C201_001_Gerber.GTO	Top Overlay (Top Silk)
20201215-VSI-Acme_CPU_Control_Board_C201_001_Gerber.GTS	Top Solder (Solder Mask)
20201215-VSI-Acme_CPU_Control_Board_C201_001_Gerber.GBS	Bottom Solder (Solder Mask)
20201215-VSI-Acme_CPU_Control_Board_C201_001_Gerber.GBO	Bottom Overlay
20201215-VSI-Acme_CPU_Control_Board_C201_001_Gerber.GM12	Board Outline
20201215-VSI-Acme_CPU_Control_Board_C201_001_Gerber.GD1	Drill Drawing
20201215-VSI-Acme_CPU_Control_Board_C201_001_Gerber.GG1	Drill Guide
20201215-VSI-Acme_CPU_Control_Board_C201_001_Gerber.GM14	Top Assembly
20201215-VSI-Acme_CPU_Control_Board_C201_001_Gerber.GM19	Bottom Assembly

Stencils (.\\Manufacturing\\Stencil Files)

Filename	Contents
20201215-VSI-Acme_CPU_Control_Board_C201_001_Stencil.GTP	Top Paste
20201215-VSI-Acme_CPU_Control_Board_C201_001_Stencil.GBP	Bottom Paste

Drill and milling data (.\\Manufacturing\\Gerber Files)

Filename	Contents
20201215-VSI-Acme_CPU_Control_Board_C201_001_NC_Drill-RoundHoles-NonPlated.TXT	NCDrill Non-plated Round Holes File
20201215-VSI-Acme_CPU_Control_Board_C201_001_NC_Drill-SlotHoles-Plated.TXT	NCDrill Plated Slot Holes File
20201215-VSI-30041113_Acme_CPU_Control_Board_C201_001_NC_Drill-RoundHoles-Plated.TXT	NCDrill Plated Round Holes File

Text Reports

Filename	Contents
20201215-VSI-Acme_CPU_Control_Board_C201_001_Gerber.REP	Gerber Generation report
20201215-VSI-Acme_CPU_Control_Board_C201_001_Gerber.EXTREP	Gerber File Extension Report
20201215-VSI-Acme_CPU_Control_Board_C201_001_NC_Drill.DRR	NCDrill File Report
20201215-VSI-Acme_CPU_Control_Board_C201_001_Stencil.REP	Stencil Generation report
20201215-VSI-Acme_CPU_Control_Board_C201_001_Stencil.EXTREP	Gerber File Extension Report

Schematic (.\\Design\\Schematic - pdf)**Assembly Drawings (.\\Assembly\\PCB Layout - pdf)**

20201215-VSI-Acme_CPU_Control_Board_C201_001_PCB.PDF

20201215-VSI-Acme_CPU_Control_Board_C201_001_DWG.PDF

Pick and Place (.\\Assembly\\Pick_n_Place)

BG95-M3_Global_TFT_SCREEN\\20201215-VSI-Acme_CPU_Control_Board_C201_001_Pick_and_Place.txt

BOM (.\\Assembly\\BOM - Excel)

BG95-M3_Global_TFT_SCREEN\\20201215-VSI-Acme_CPU_Control_Board_C201_001_BG95-M3_Global_TFT_SCREEN_DFM_BOM.xls

Additional info**Mechanical Files (.\\Assembly\\Mechanical)**

No Variations\\20201215-VSI-Acme_CPU_Control_Board_C201_001_DXF_Model.DXF

BG95-M3_Global_TFT_SCREEN\\20201215-VSI-Acme_CPU_Control_Board_C201_001_BG95-M3_Global_TFT_SCREEN_3D.step

General

Extended gerber / excellon drill.

Layout

Filename: Altium 17.1: Control_Board_Model_C201_FFC_Sens_Conn.PcbDoc

Fiducials

TOP FID1: 2.54mm; 62.865mm

TOP FID2: 2.54mm; 7.62mm

TOP FID3: 93.98mm; 6.35mm

BOT FID1: 2.54mm; 62.865mm

BOT FID2: 2.54mm; 7.62mm

BOT FID3: 93.98mm; 6.35mm

Contacts

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